

WHO IS HE?!? Your **TA** Robin Carpenter – Senior in Physics

22yo, from Wisconsin, USA

“I don’t know how I ended up here but I
sure hope I can leave too”

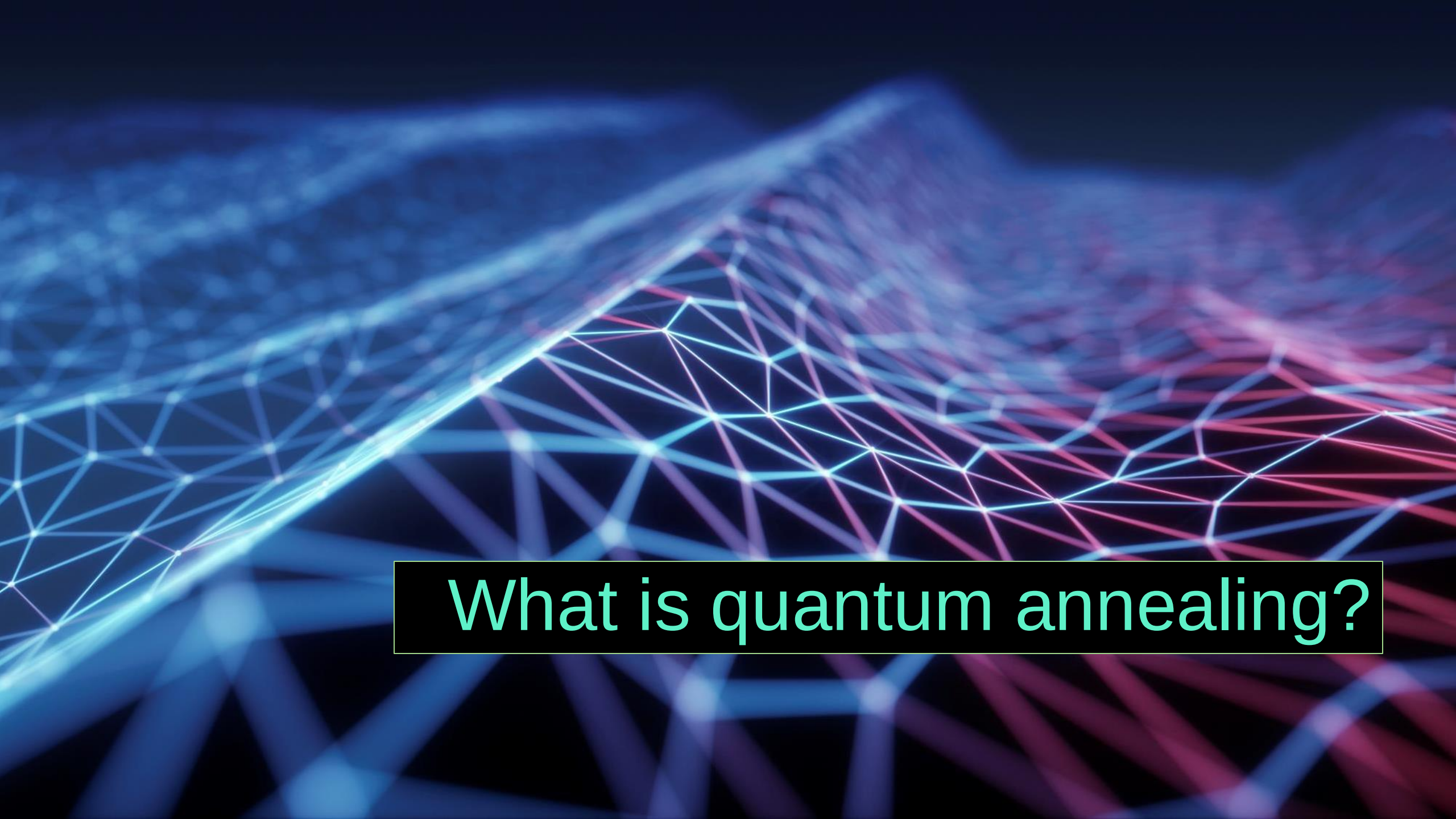
Have a personal passion for quantum
information science



■ This section - **QUANTUM COMPUTING**

- Focusing on the DWAVE quantum annealer.
- Our previous projects are on vertex clustering
 - I would like to change or adapt this topic
- **HW every other week**, will send a schedule soon. Each homework is a group activity and relatively easy
- Weekly meetings for groups! Outside of class time
- Annnnd ... We need to start weekly presentations from teams. This is valuable experience

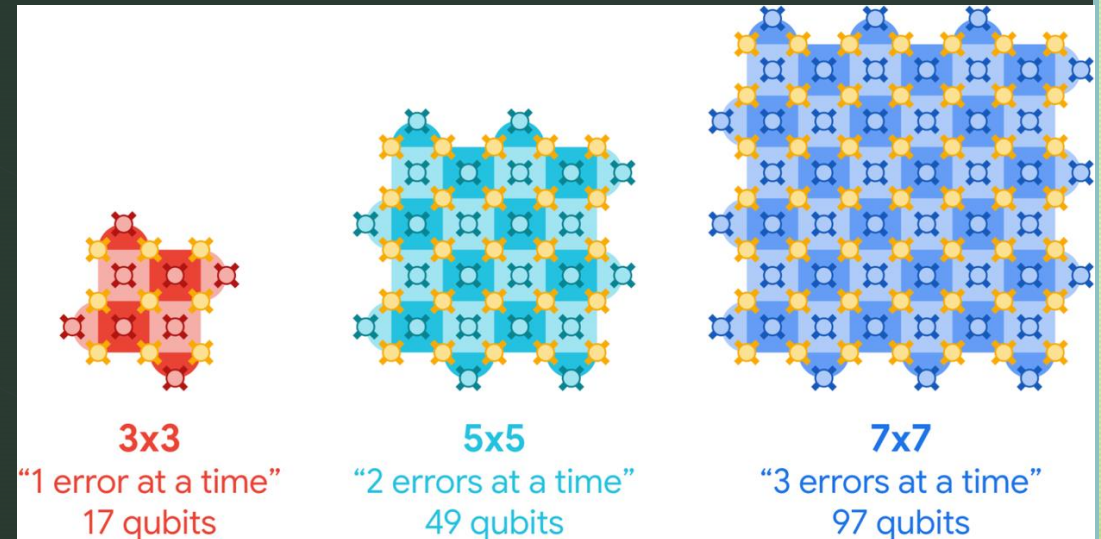
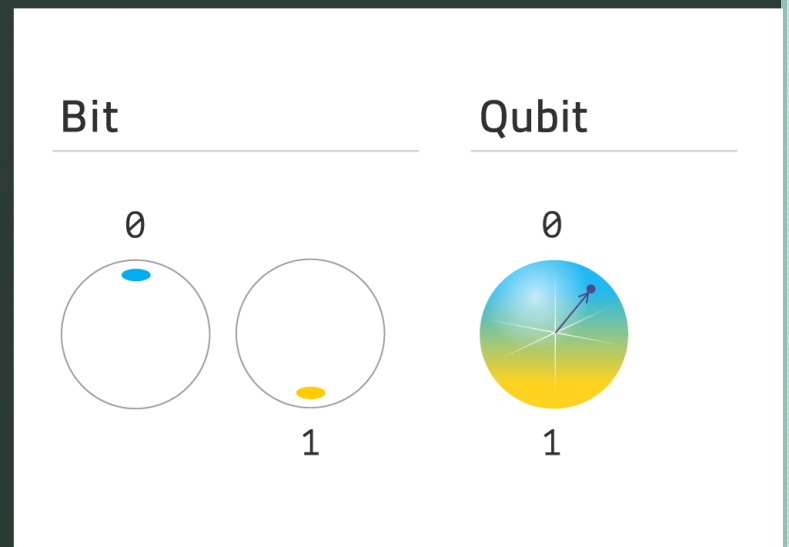




What is quantum annealing?

QUANTUM COMPUTERS :::

- Proposed around 50 years ago
- Have been realized with different information bases (qubits of light, atoms, currents)
- Have a LOT of error



Quantum computing – cold and very difficult

QUANTUM ANNEALING IS THE "LITTLE COUSIN" OF QC



Annealing isn't bad or dumb, it is a clever solution ... to not many conceived practical problems.

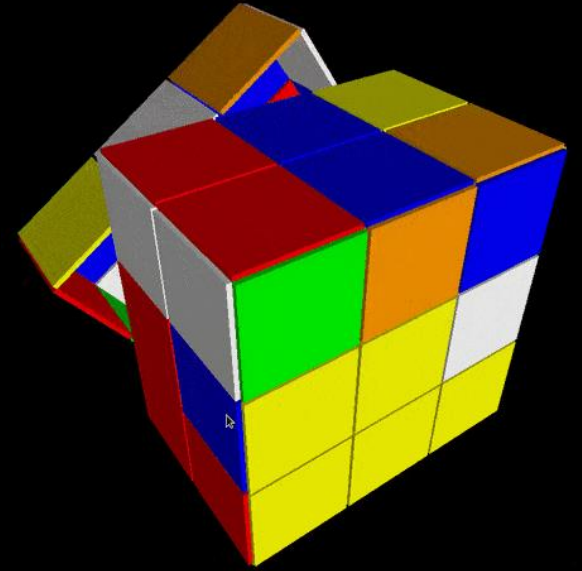
What makes annealing unique?

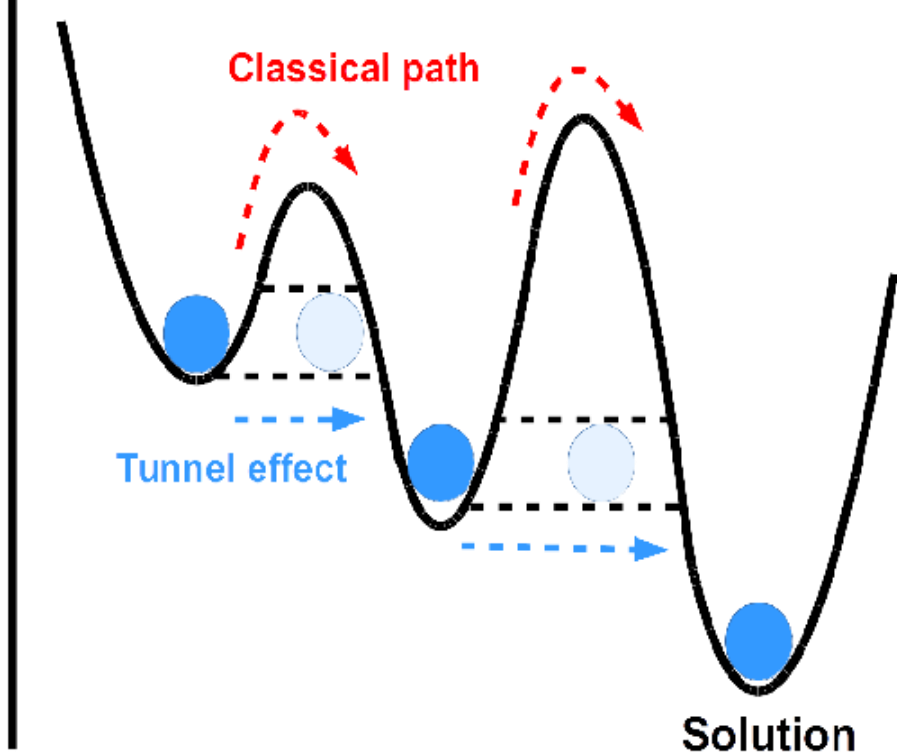
Using physics to optimize

QA is very well suited for combinatorics, where **every possible** solution of a problem is probed to find the best.

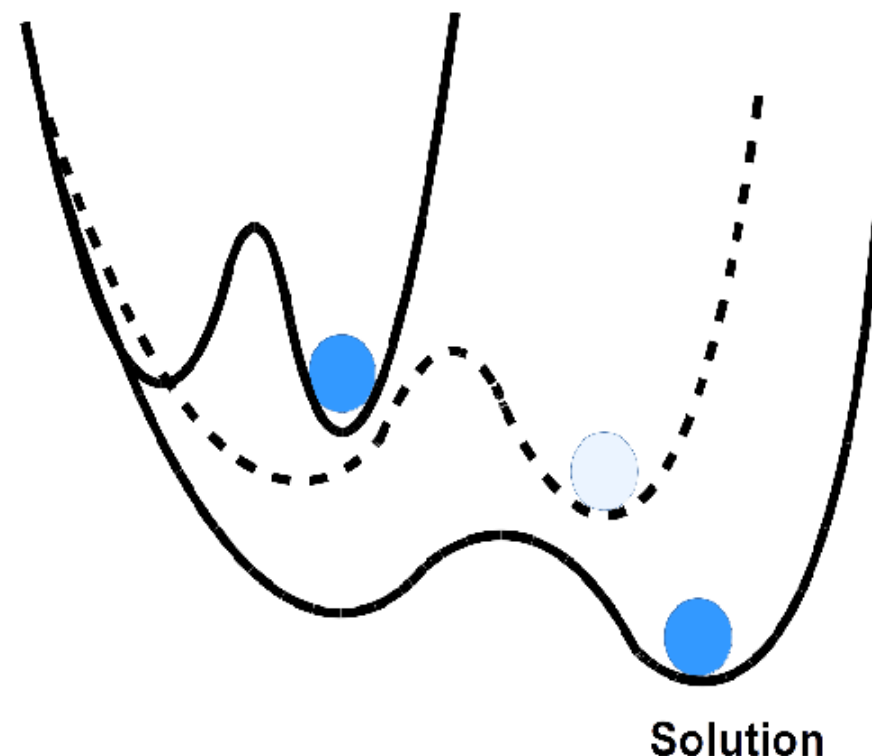
By doing the computation slow enough, quantum tunneling and the adiabatic theorem will ensure that the best solution is found.

Keep in mind – this exact process can be simulated on computers already!





Quantum Tunnelling

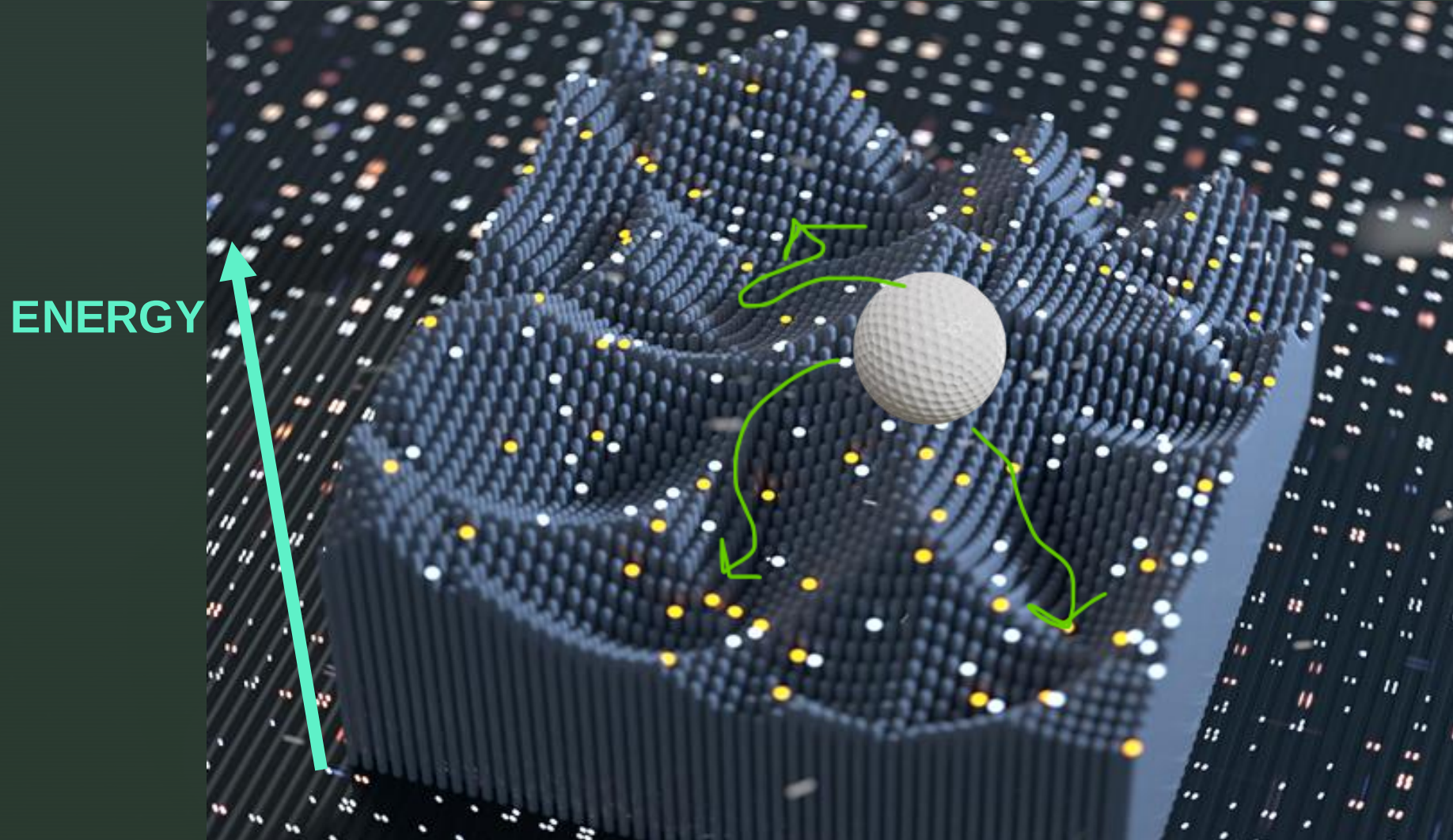


Adiabatic evolution

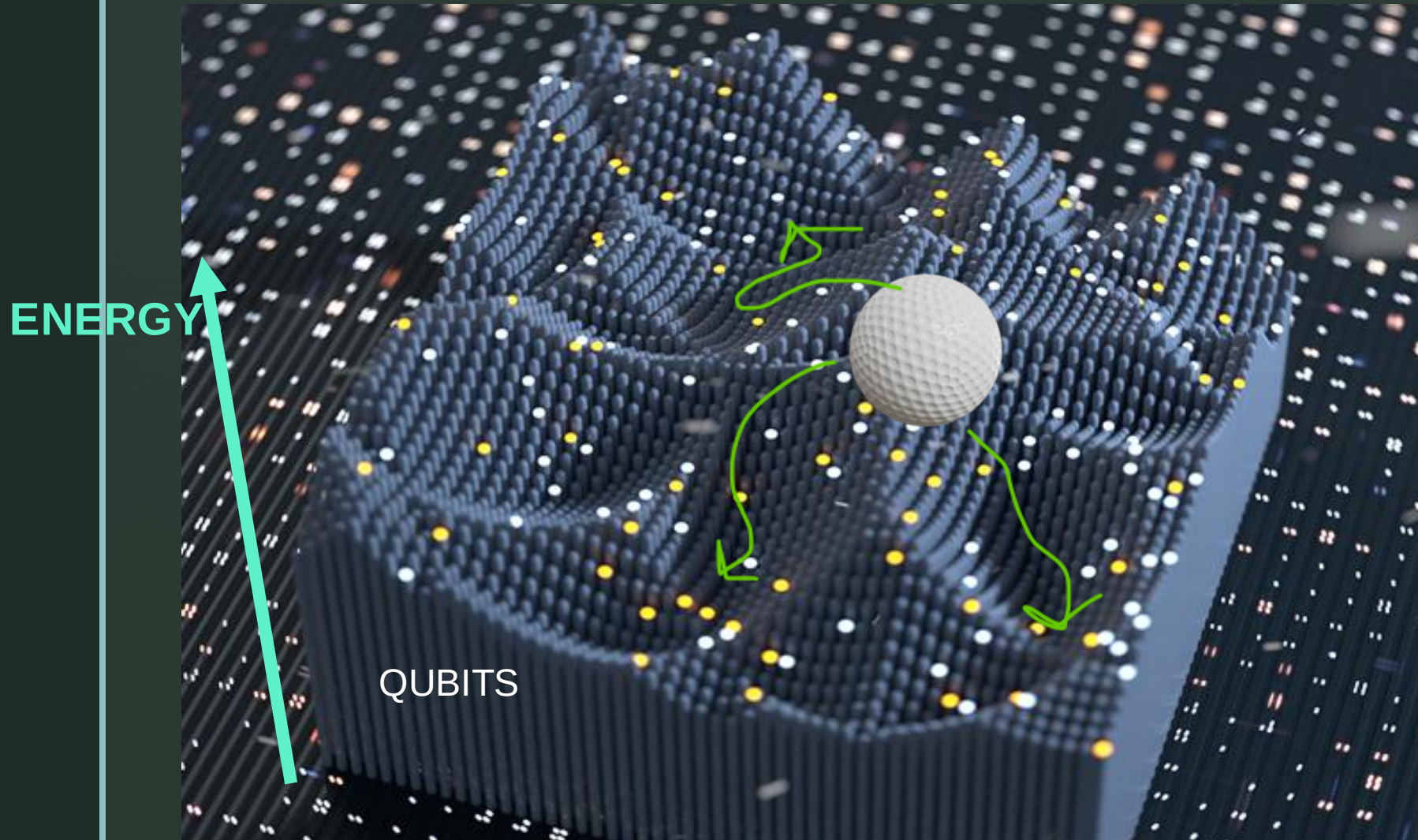
Essentially an “adiabatic solver”

How does an annealer work?

Every quantitative problem has a space of solutions – leveled by cost



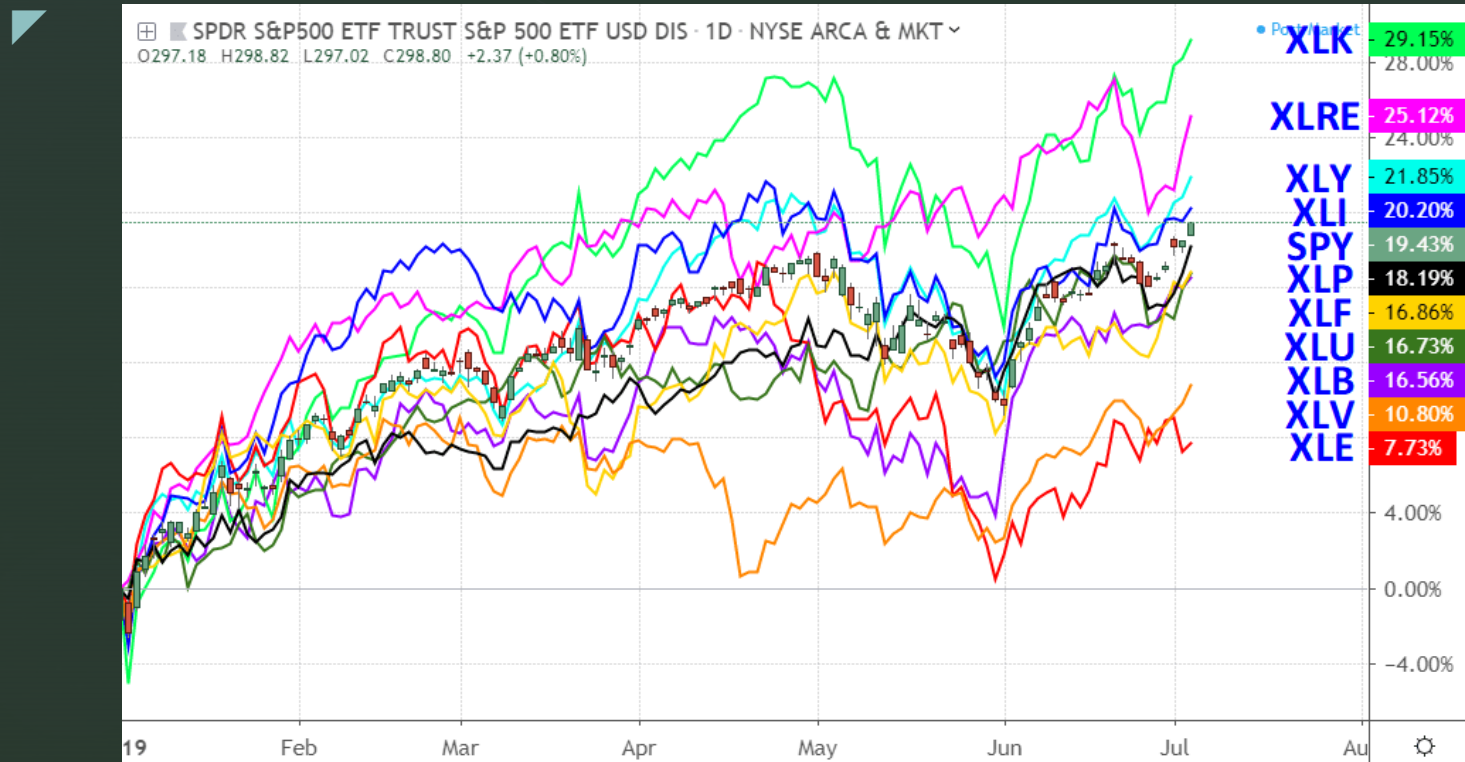
Using quantum mechanical bits, controlling the system *slowly* finds the optimal solution with highest probability



What are its best uses?

Anything involving rugged “energy
landscapes.”

Finance / Molecular simulation / logistics



Examples continued ...

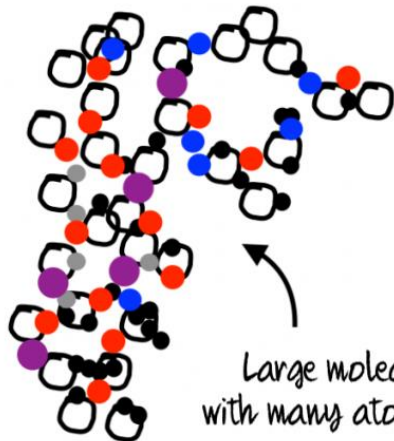
traveling salesman problem, transportation, quantum chemistry, materials science, simulating quantum systems

What computers can simulate today



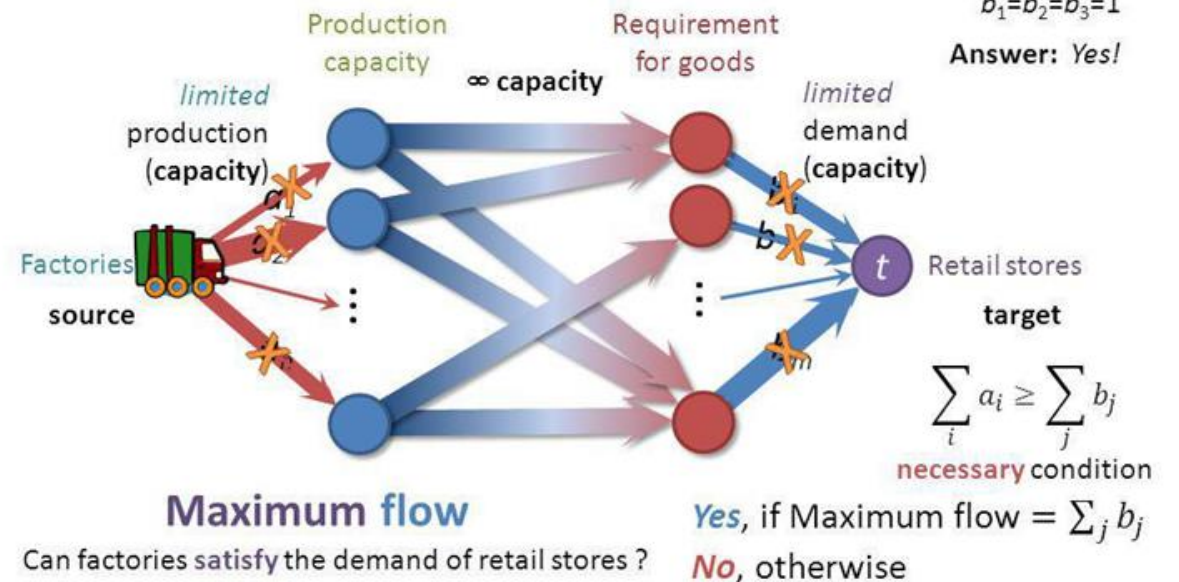
Small molecule with a few atoms

What quantum computers could simulate



Large molecule with many atoms.
I don't know, maybe a protein or something

• Transportation problem

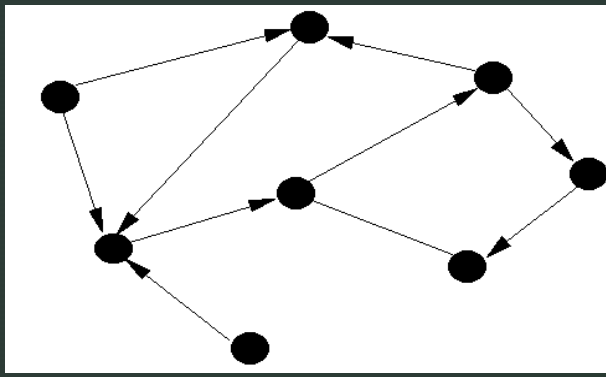


Conceptual example: Finance

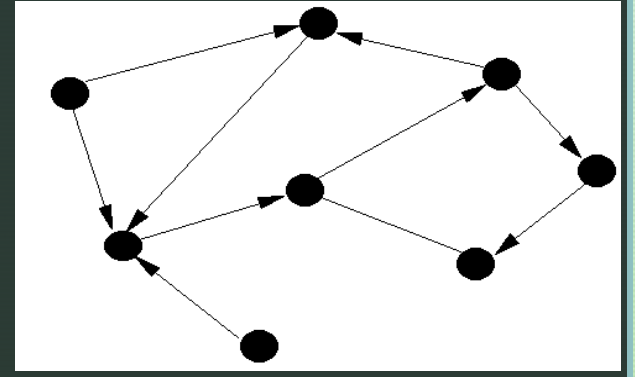
Make the most money in stocks today, given...

1. I only have 500\$
2. Choice between 100 stocks
3. I can only invest in N stocks
($N = 1, 2, 3, \dots$)
4. All relevant stock parameters
(open, high, low, volume ...)

Microsoft Corp \$46.41 +0.37	
Last trade	46.41
Trade time	Sep 26 04:00 PM Eastern Daylight Time
Change	0.37 (0.80%)
Prev close	46.04
Open	45.93
Bid	46.39
Ask	46.40



OK, but how?



THE ISING SPIN EQUATION

$$H = \text{Energy of system} = - \sum_{i,j} J_{ij} \sigma_i \sigma_j + h \sum_i \sigma_i$$

What do we see here?

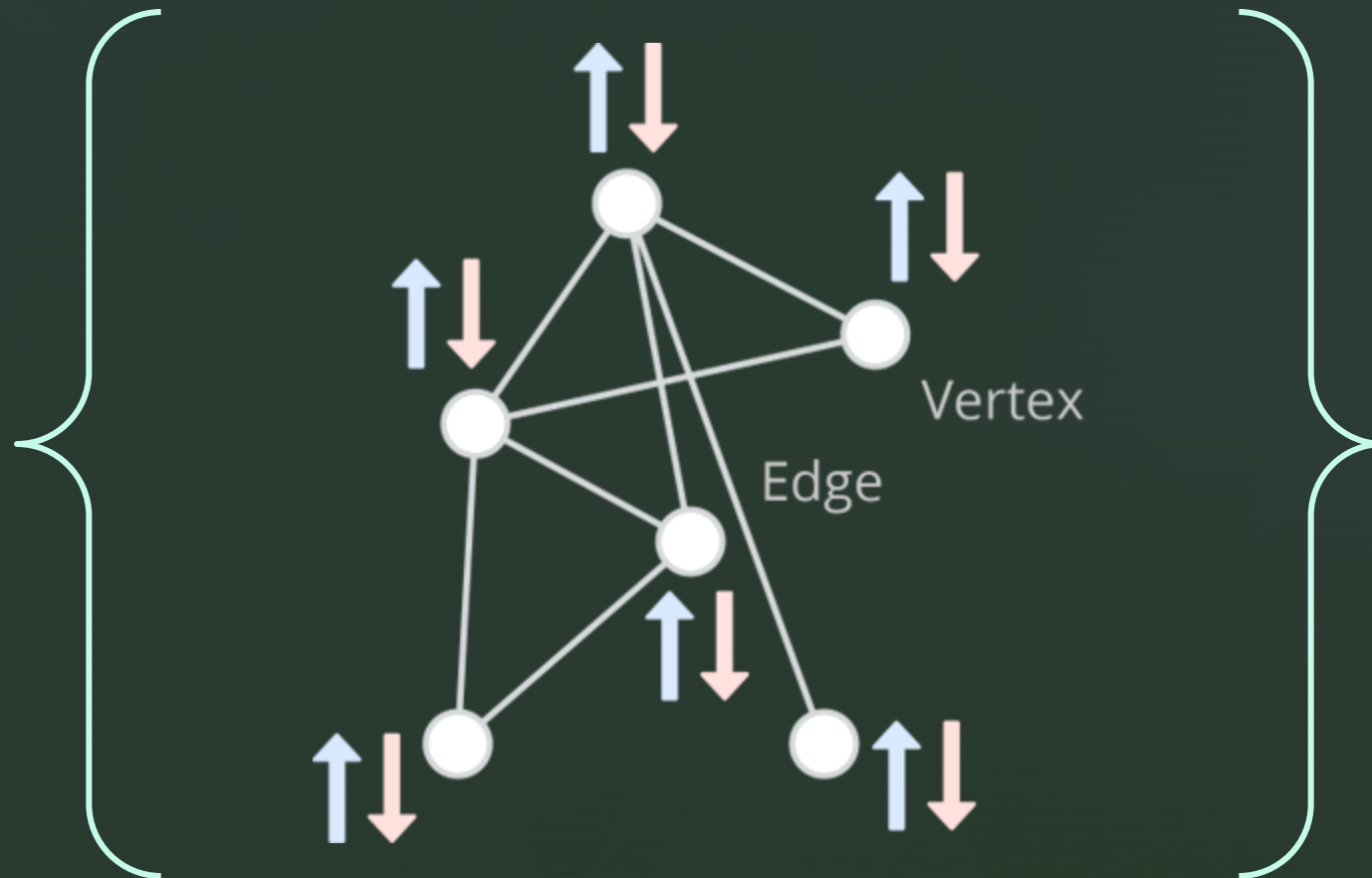
Why two summations?

Why i and j? What do they mean?

Summation 1:
considers all of
the couplings
between each
qubit (edges)

$$H = - \sum_{i,j}^1 J_{ij} \sigma_i \sigma_j + h \sum_i^2 \sigma_i$$

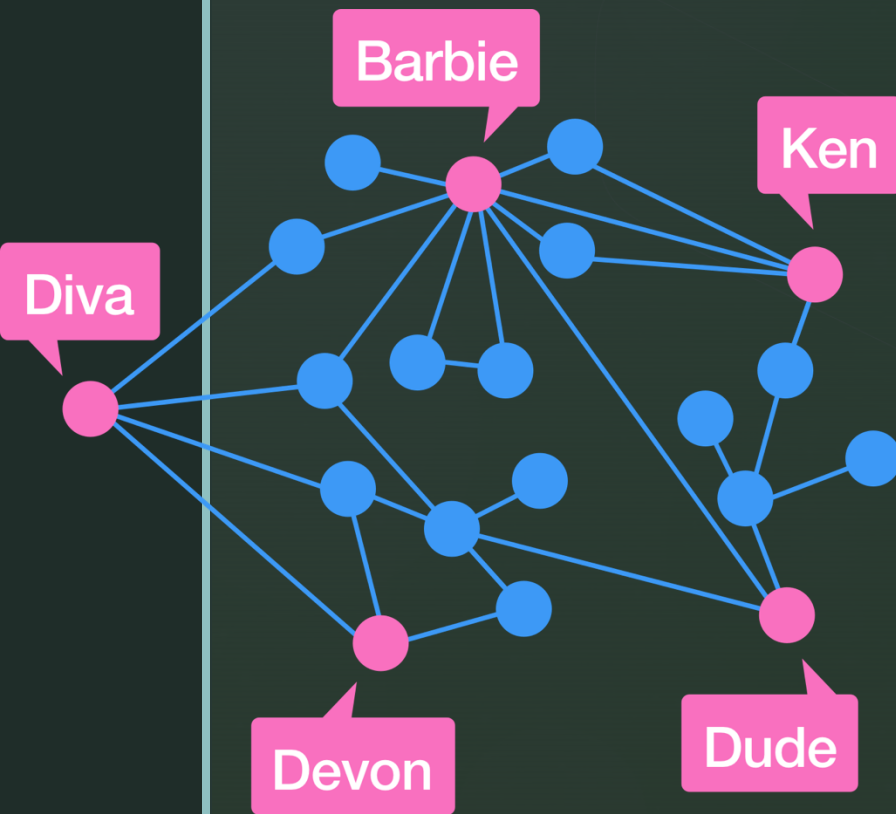
Summation 2:
considers each
individual bias of
every qubit
(graph vertexes)



That's it – we encode the entire problem into the Ising equation!

- Quite powerful – any problem can be broken down into a simple network. Just unknown elements interacting with each other knowingly.
- Keep in mind, a perfect graph has full connectivity – each qubit links to every other.
 - $4 \mid 3+2+1 = 6$
 - $6 \mid 5+4+3+2+1 = 15$
 - $10 \mid 9+8+7+6+5+4+3+2 = 45$
- As n increases, number of connections m approaches $\frac{1}{2} * n^2$

Example: social network

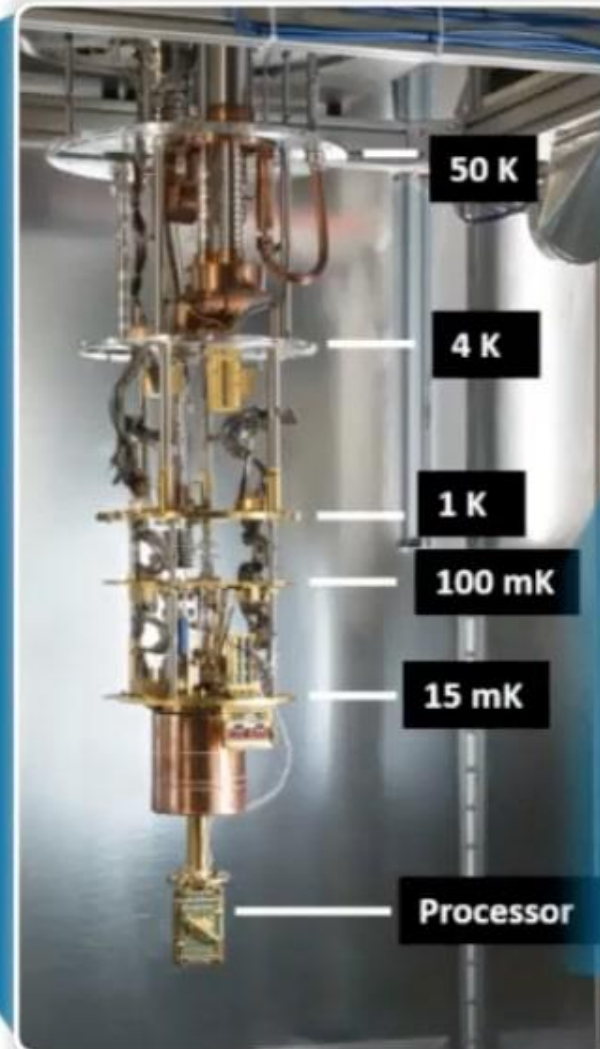


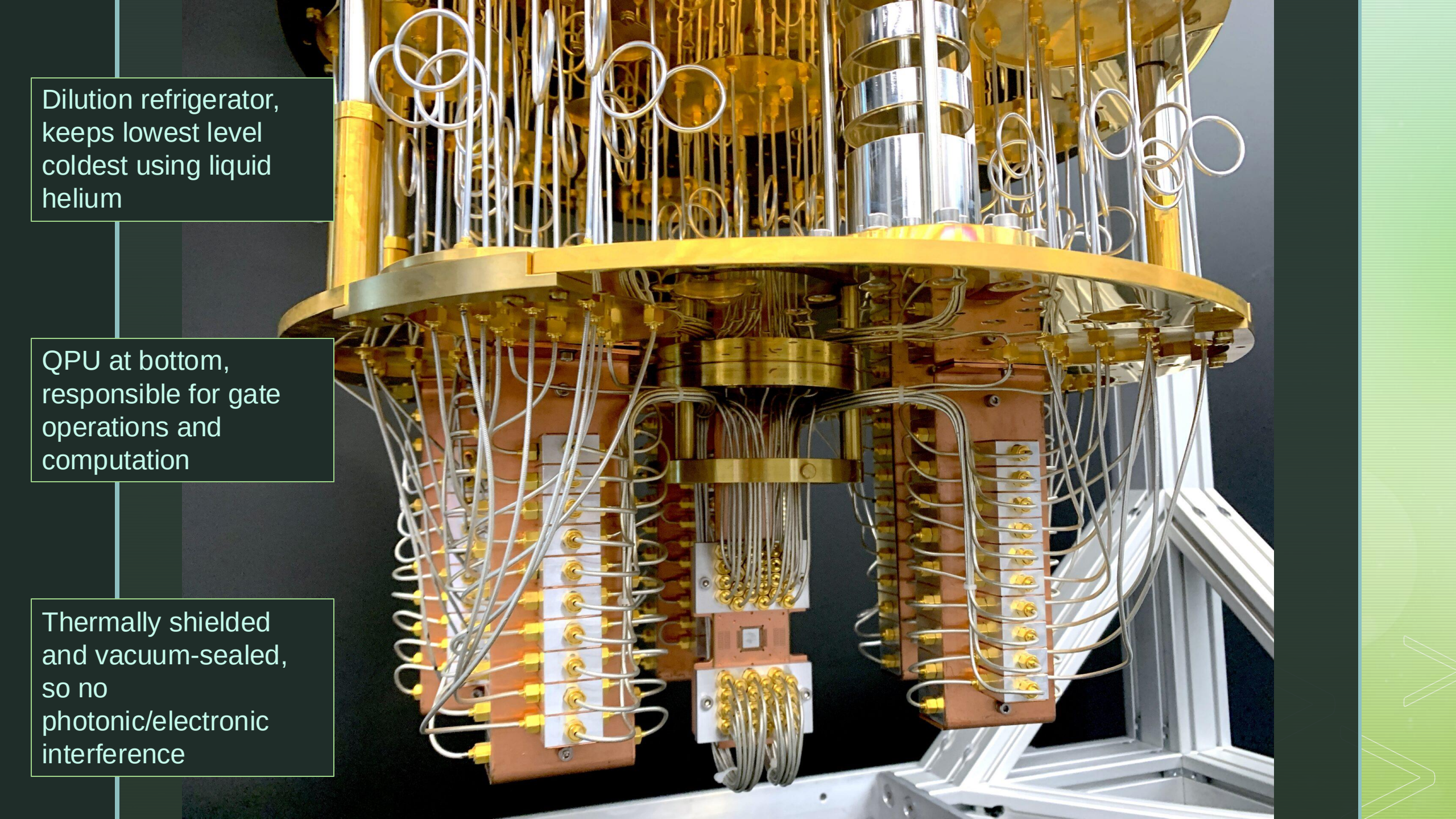
1. Define problem. I want to choose friends (1s) and enemies (0s) from a random group of people.
2. I. Biases: how much I agree with someone.
3. Couplings: how much people agree with each other.
4. Now do it!

$$N=5, H = -([\sigma_0 - 2\sigma_1 + 2\sigma_2 - 4\sigma_3 + \sigma_5] + [2\sigma_0\sigma_1 - 10\sigma_0\sigma_2 + \dots])$$

After coding in QA, each σ will become a 0 or a 1, and the optimal solution is where I have friends with high coefficients and enemies with negative coefficients.

What Is A Quantum Computer



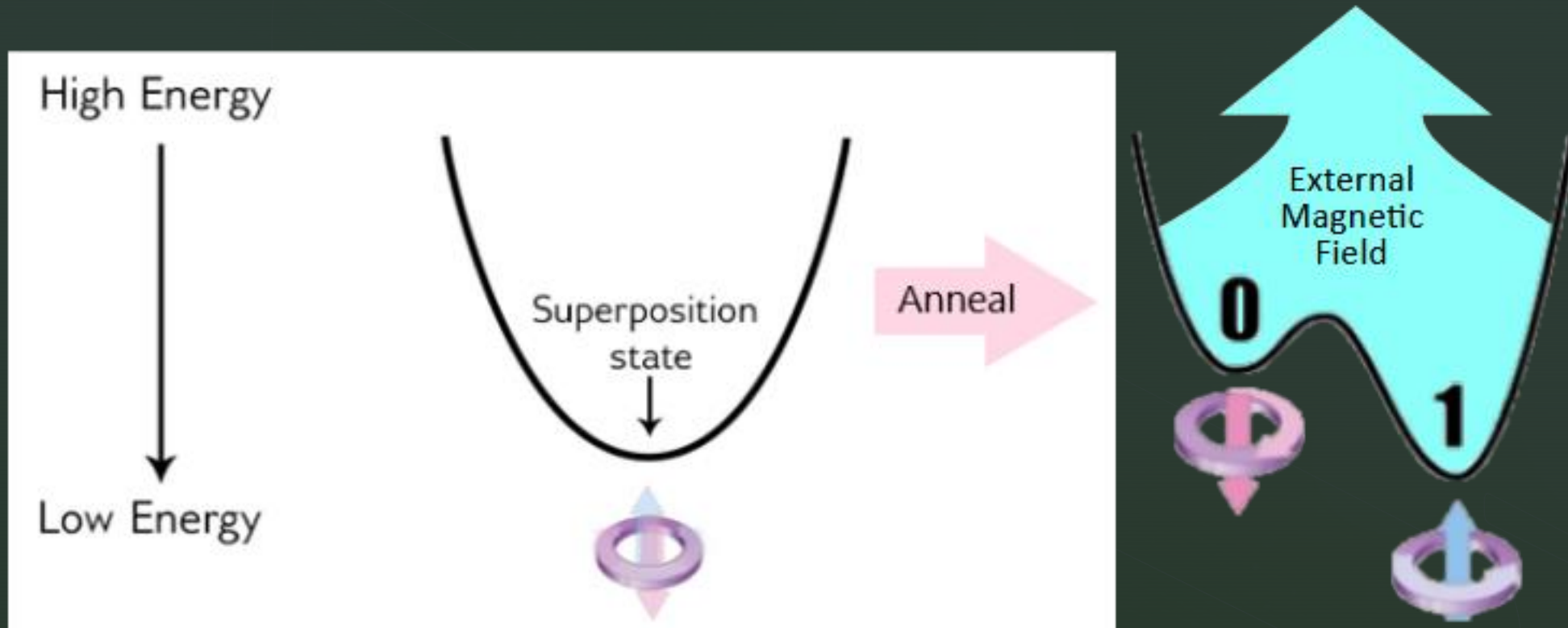


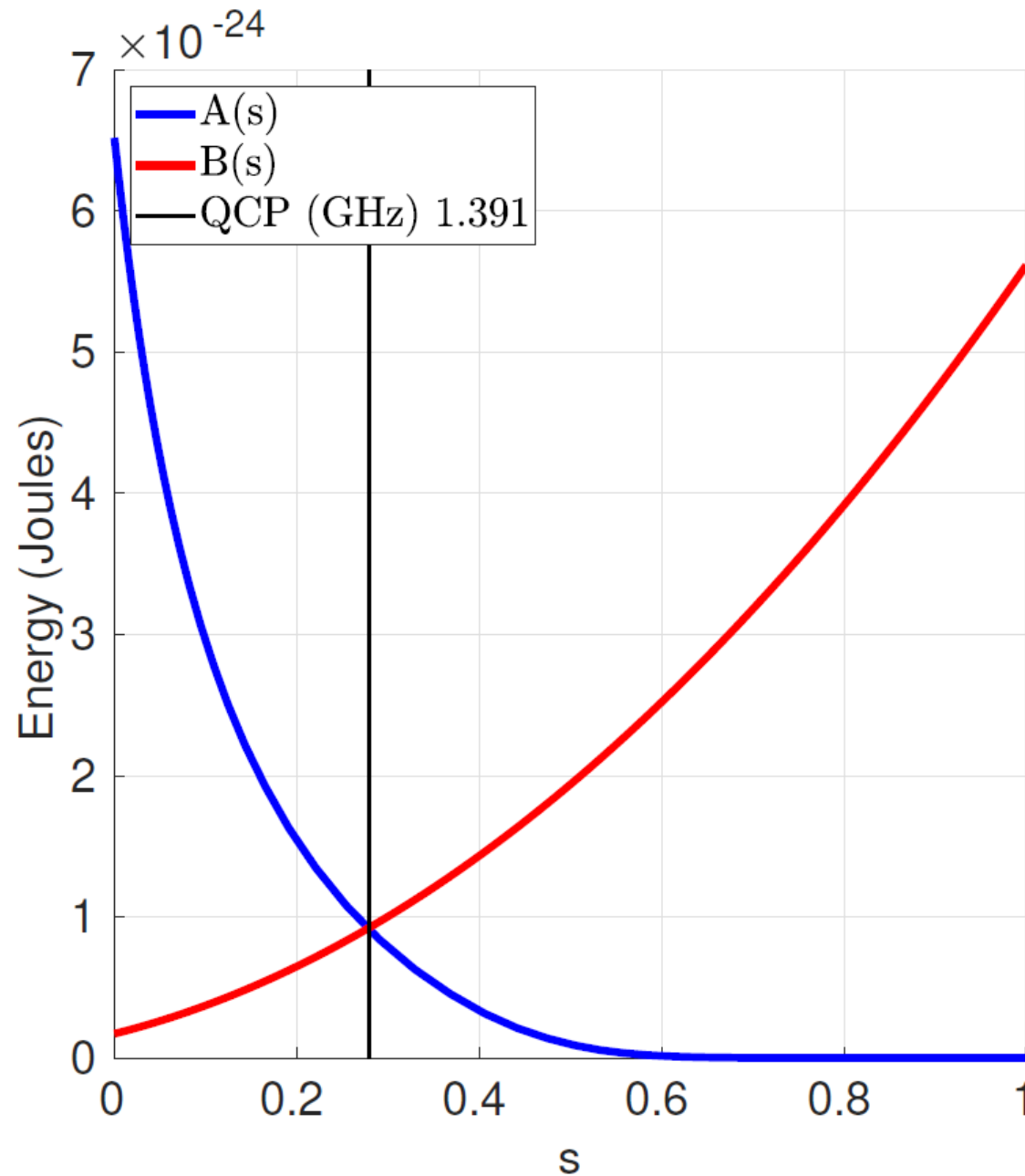
Dilution refrigerator,
keeps lowest level
coldest using liquid
helium

QPU at bottom,
responsible for gate
operations and
computation

Thermally shielded
and vacuum-sealed,
so no
photonic/electronic
interference

- Ok, how does a DWave processor work?



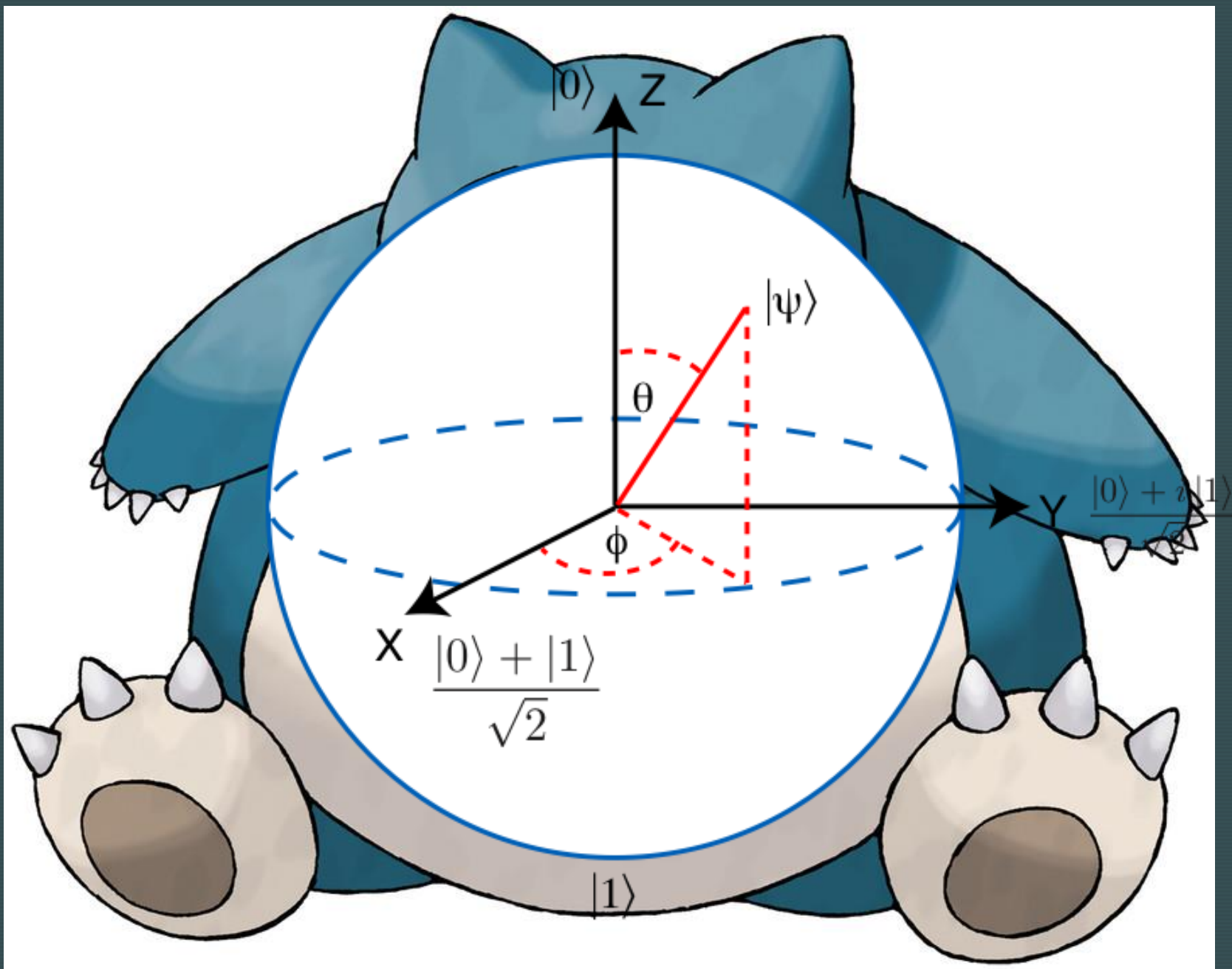


$A(s)$ = initial configuration. Each qubit is superimposed, some default setting of biases. No couplings needed!

$B(s)$ = problem configuration. Qubits literally quantum tunnel from $A(s)$ to $B(s)$ given a slow change (adiabatic theorem).

S – think of it like time, normalized to go from 0 to 1.

SNUBIT





Universal QC (IBM, google, ...)

- Use quantum gates, can model ANY classical algorithm
- Largest processors ~ 120 qubits

DWave's QA

- Just uses a coupled graph for binary optimization. Cannot model any classical algorithm
- Largest processors ~ 5000 qubits